

t-Closeness: Privacy Beyond k-Anonymity and l-Diversity

Summary

The article introduces a new concept of t -closeness which is a privacy model for data publishing. First however, the article reviews key concepts of k -anonymity, which refers to the uniqueness of the quasi identifiers, and l -diversity, which refers to the uniqueness of the sensitive values within their groups. However the authors note a key flaw with l -diversity which is that the distribution of the sensitive attribute within subgroups. As such, they proposed the idea of t -closeness which is the idea that the distribution of the sensitive attributes within a subgroup is similar to the distribution in the overall table.

Strengths

- k -anonymity as a simple identity concealing metric
- l -diversity as a direct combatant to homogeneity attacks
- t -closeness direct adressment of the limits of l -diversity

Weaknesses

- k -anonymity lack of protection against sensitive attributes
- l -diversity insufficient protection of sensitive attributes
- t -closeness computational complexiity
- t -closeness weakens the researching utility of the data

Comments On Strengths & Weaknesses

The idea of k -anonymity is that it ensures the identity of individuals are concealed within the data set by ensuring there is at least k matching quasi identifiers for any set of quasi-identifiers in the data set. However, this is subject to homogeneity attacks as if a particular group of quasi-identifiers has a low variety of the sensitive attribute within the group. For example, even if we achieve k -anonymity for a electronic health record dataset, if a group only has heart related issues then an attacker could easily determine that a person in that group has a heart related issue. This gets addressed by l -diversity which ensures that knowledge about a person's group does not directly give away their sensitive attribute. With distinct l -diversity, it ensures there are at least l different values for the sensitive attribute within a given group. This however is not necessarily sufficient as a single entry of a different value would achiheve l diversity although it would not add much protective power. This leads to other definitions of l diversity such as the entropy definition but this too has draw backs if the table overall has a skewed distribution. This leads us to the t -closeness definition which aims to have the distribution of values within a group match the distribution of the overall table within a threshold t . This of course could be quite computationally expensive especially if there are multiple sensitive attributes each with different distributions. Additionally, this is a direct decrease of the utility of the data since a well-intentioned researcher is less able to utilize the data to draw distinctions of the sensitive attribute across different quasi-identifiers.